

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Computer Science and Engineering) SEMESTER - 2 Summer 2025 (Regular)

Course :Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 2

Subject Code & Name: 24AF1EMES204 - ENGINEERING MECHANICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1 A uniformly distributed load (UDL) acts over:

- a) A single point
- b) A specific length with uniform intensity
- c) The entire structure with varying intensity
- d) A concentrated section of the structure

In a free body diagram, the weight of the object is usually represented as acting:

- a) Horizontally
- b) At the centroid vertically downward
- c) At the support points
- d) Tangentially

3 When two or more forces acting at a point are combined into a single force, it is called:

- a) Equilibrant
- b) Resultant force
- c) Couple
- d) Load factor

4 A body is in static equilibrium if:

- a) The resultant force and resultant moment are both zero
- b) The resultant force is positive
- c) The moment is greater than zero
- d) The net force equals mass times acceleration

5 The centroid of a rectangle is located at:

- a) One-fourth of the height and one-fourth of the width
- b) At the intersection of diagonals
- c) At the midpoint of any side
- d) One-third of the height from the base

- 6 When a body is just about to move, the frictional force acting on it is called:
- Sliding friction
 - Rolling friction
 - Limiting friction
 - Dynamic friction
- 7 In a Hinged or Pinned Support, how many reactions are developed?
- Two
 - Three
 - Four
 - Six
- 8 In the formula for truss analysis $m = 2j - 3$, the variable 'm' stands for:
- Number of supports
 - Number of members
 - Number of joints
 - Number of reactions
- 9 If an object is thrown upward with an initial velocity u , the velocity at the highest point is:
- Equal to u
 - Zero
 - Infinite
 - Equal to acceleration
- 10 According to D'Alembert's principle, the equation of motion can be written as:
- $F - ma = 0$
 - $F + ma = 0$
 - $F = ma$
 - $F = m/v$
- 11 The principle of work and energy for a rigid body in translation states that:
- Work done = Change in kinetic energy
 - Work done = Change in potential energy
 - Work done = Force $\times$ Distance
 - Work done = Momentum $\times$ Velocity
- 12 The kinetic energy of a moving object depends on:
- Velocity and mass
 - Acceleration and mass
 - Force and displacement
 - Work and power

Q2. Solve the following.

- State and explain Newton's three laws of motion. Enlist idealisations made in engineering mechanics. 6
- State and prove Varignon's theorem. 6

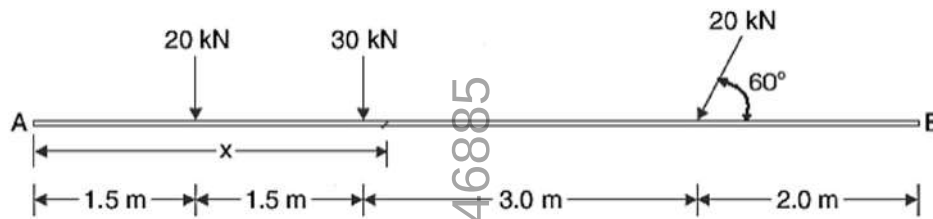
Q3. Solve the following.

- An I-section has the following dimensions in mm units: 6
 Bottom flange = 300×100 , Top flange = 150×50 , Web = 300×50
 Determine mathematically the position of centroid of the section. Take bottom of the bottom flange be the axis of reference.

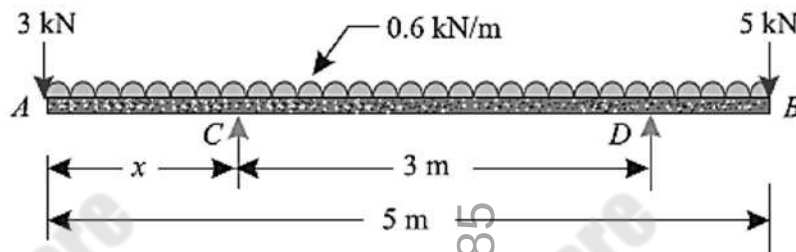
- B) A body, resting on a rough horizontal plane, required a pull of 180 N inclined at 30° to the plane just to move it. It was found that a push of 220 N inclined at 30° to the plane just moved the body. Determine the weight of the body and the coefficient of friction. 6

Q4. Solve Any Two of the following.

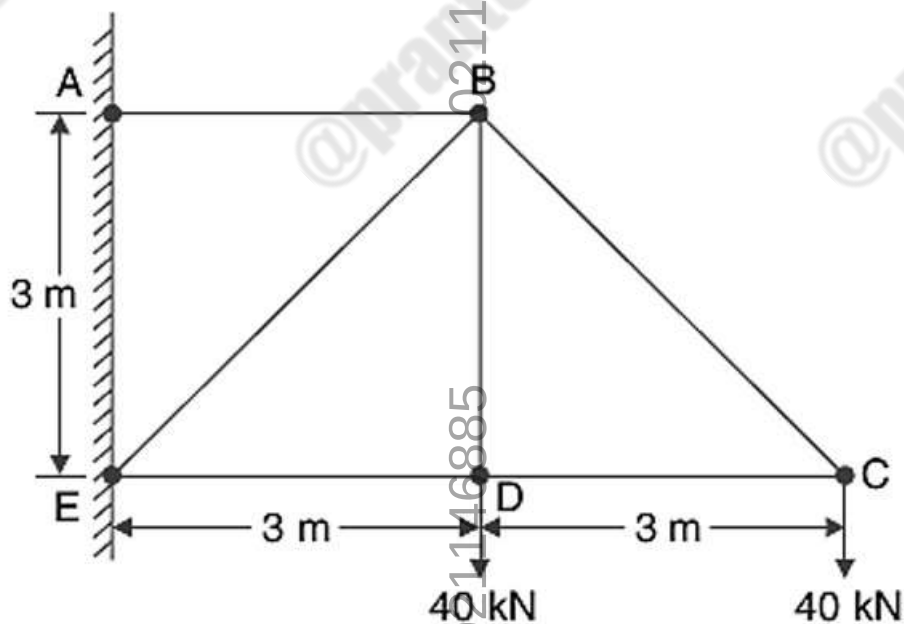
- A) Determine the resultant of the system of forces acting on a beam. 6



- B) A beam AB 5 m long, supported on two intermediate supports 3 m apart, carries a uniformly distributed load of 0.6 kN/m . The beam also carries two concentrated loads of 3 kN at left hand end A, and 5 kN at the right hand end B. Determine the location of the two supports, so that both reactions are equal. 6



- C) Find the forces in all the members of the truss shown in figure. Tabulate the results. 6



Q5. Solve Any Two of the following.

- A) A small steel ball is shot vertically upwards from the top of a building 25 m above the ground with an initial velocity of 18 m/sec. 6
 a) In what time, it will reach the maximum height?
 b) How high above the building will the ball rise?
- B) A ball is dropped from the top of a tower 30 m high. At the same instant a second ball is thrown upward from the ground with an initial velocity of 15 m/sec. When and where do they cross and with what relative velocity? 6

- C) If a particle is projected inside a horizontal tunnel which is 5 meters high with velocity of 60 m/s, find the angle of projection and the greatest possible range. 6

Q6. Solve Any Two of the following.

- A) State and prove work energy principle. 6

- B) i) Define Direct Impact, Oblique Impact 6

ii) State Impulse and Momentum Principle

iii) State Law of Conservation of Momentum

- C) A 1500 N block is in contact with a level plane, the coefficient of friction between two contact surfaces being 0.1. If the block is acted upon by a horizontal force of 300 N, what time will elapse before the block reaches a velocity of 16 m/sec starting from rest? If 300 N force is then removed, how much longer will the block continue to move? Solve the problem using impulse momentum equation. 6

*** End ***